

ODONT 541.1

CARIOLOGY

Classification

Lecture #6

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NO recommended textbook reading for classification or diagnosis

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Types of Classification

Ways that the dental system describe caries.

- GV Black
 - What are the most common sites of caries?
 - Do we see or anticipate an etiologic pattern based on these sites?
- Bitewing depth
 - How deep is it radiographically?
- Active vs Inactive
 - Is it currently growing or is it stagnant?
- Cavitated vs non cavitated
 - Is there a hole serving as a protected site?
- ICDAS
 - How well can we correlate visual appearance to histologic depth?
 - How "bad" is this carious lesion?

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GV Black Classifications

Class I: Caries affecting pits and fissures on occlusal third of molars and premolars, occlusal two-thirds of molars and premolars, and lingual part of anterior teeth.

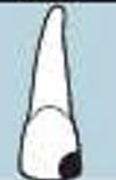
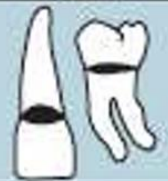
Class II: Caries affecting proximal surfaces of molars and premolars.

Class III: Caries affecting proximal surfaces of central incisors, lateral incisors, and cuspids without involving the incisal angles.

Class IV: Caries affecting proximal including incisal angles of anterior teeth.

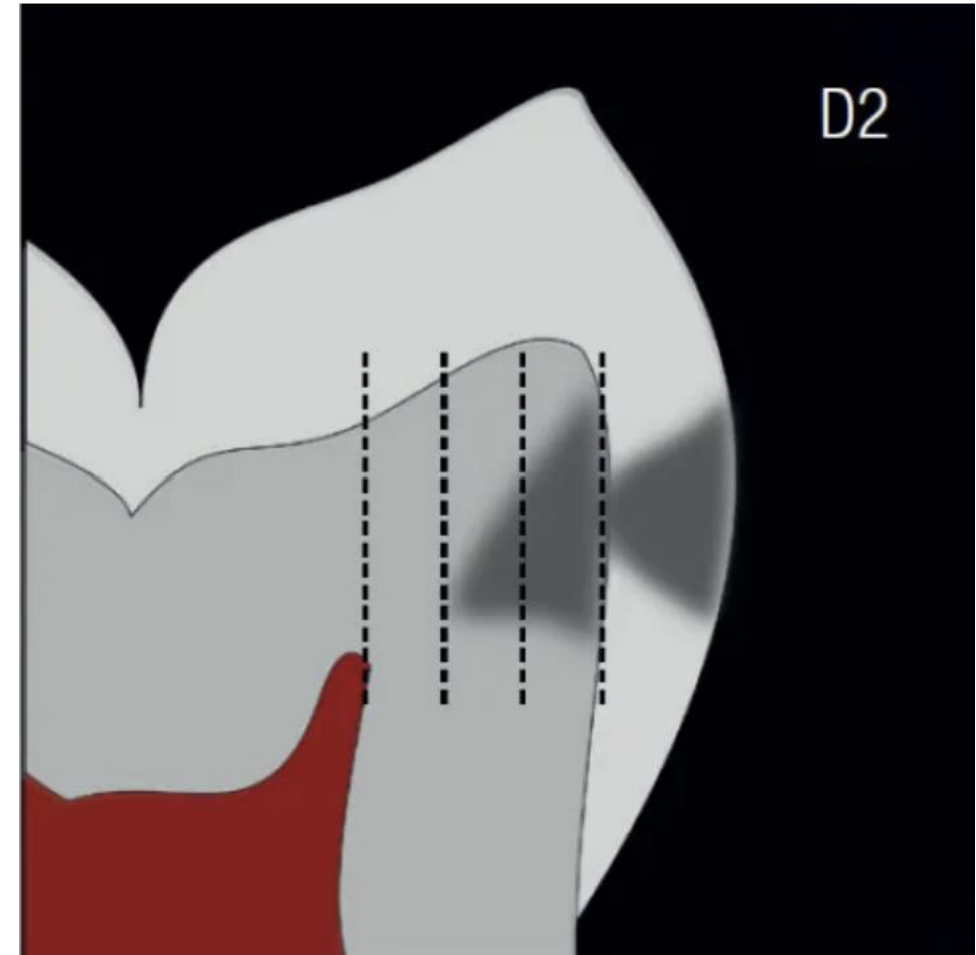
Class V: Caries affecting gingival one-third of facial or lingual surfaces of anterior or posterior teeth.

Class VI: Caries affecting cusp tips of molars, premolars, and cuspids

Classes	Illustration
Class I	
Class II	
Class III	
Class IV	
Class V	
Class VI	

Divide the depth of the enamel from the outer surface to the DEJ into halves.

Divide the depth of the dentin from the DEJ to the pulp into thirds.

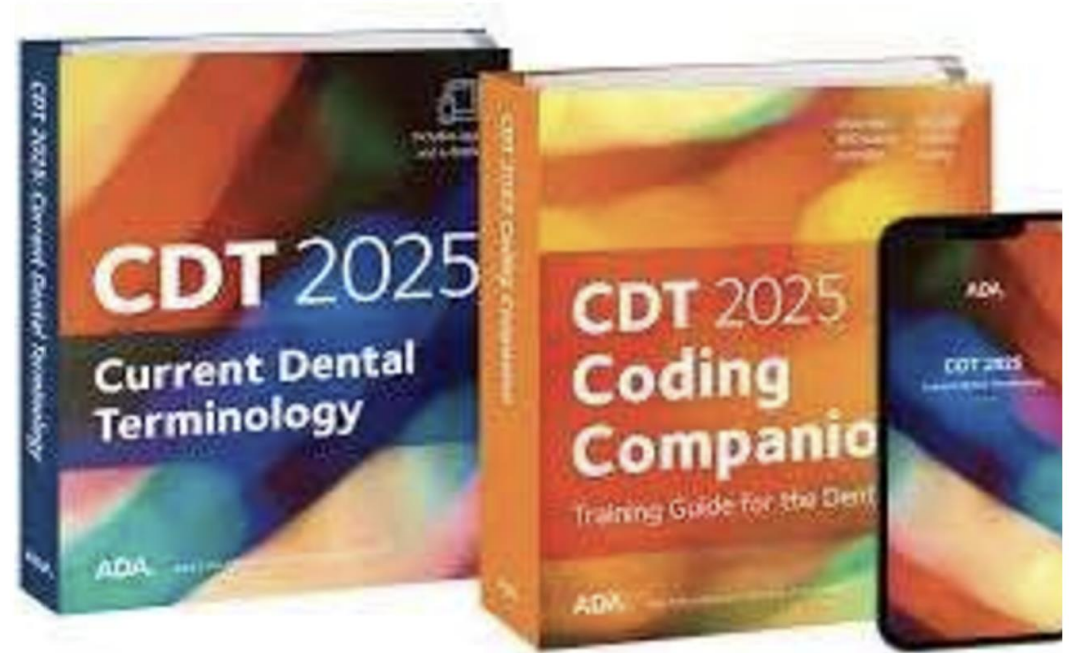


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CDT Classification

- "Current Dental Terminology" code book updated by the ADA annually.
- CDT coding has no scientific or medical basis. It classifies **treatments** of all kinds (restorative, orthodontics, surgical, etc) into a common language for billing purposes to insurance companies.
 - D0XXX is for diagnostic
 - D1XXX is for preventive
 - D2XXX is for restorative (direct and indirect)
 - D3XXX is for endodontics
- CDT coding is **ONLY** required for insurance communication
 - If no insurance involved, use of CDT is not required.
 - However, if you choose to use CDT (even without insurance), your coding must be truthful to avoid fraud.
 - Codes are frequently added, deleted, or redefined.



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CDT Codes

- D2330:** Resin-based composite – 1 surface, anterior
- D2331:** Resin-based composite – 2 surfaces, anterior
- D2332:** Resin-based composite – 3 surfaces, anterior
- D2391:** Resin-based composite - 1 surface, posterior
- D2392:** Resin-based composite - 2 surfaces, posterior
- D2393:** Resin-based composite - 3 surfaces, posterior

- You do not need to memorize any CDT codes for this course. But if given a list of CDT codes with descriptions (including ones not listed on this slide), you should be able to match the code to the restoration.
- A 2 surface composite can be a class 1 (OL), a class 2 (MO), a class 3 (ML), a class 5 (DF).
- Which code would you use to bill insurance for a #8MLF composite?
- Does the insurance company care if your restoration was big, small, difficult, took excessive time, or of exceptionally high quality? NO. What if you care about those things?
- It is the dentist's responsibility to ensure any coding being used is correct and up to date.
 - Practice management programs, such as Axium or Dentrix, do not automatically update their codes annually.
 - If an office manager (or employer dentist) miscodes your work, you are still responsible.

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Cavitated vs Not Cavitated

"A distinctive stage in the caries process is the stage of cavity formation. When a carious cavity has formed it is much more difficult to control the biofilm by oral hygiene procedures. Therefore, the treatment of choice for such cavitated lesions usually involves operative intervention in the form of restorations. This intervention **will not manage the causes of caries, but restoring the tooth makes it easier to perform proper oral hygiene.**"

Active vs. Inactive

We can assess if a carious lesion should be treated as active or inactive based on several factors:

- Location: Is it in a plaque stagnation area?
 - o **If a lesion is covered in plaque, it is presumed active**
- Appearance:
 - o Active: Matte, warm colors
 - o Inactive: Shiny, cold colors
- Texture
 - o Active: Rough
 - o Inactive: Smooth
- Integrity
 - o Cavitated: even if arrested, likely to reactivate. Treat as active
 - o Non cavitated: even if active, likely to be successfully arrested. Consider non invasive treatment if no dentinal involvement.
- Bleeding on probing:
 - o Active: If the gingiva bleeds, this suggests the recent presence of plaque. Did your patient brush just for their dental visit?
 - o Inactive: Gingival health suggests consistent plaque removal
- Risk Factors
 - o High cariogenic diet, significant xerostomia
 - Unlikely to maintain in active state unless risk factors can be altered

	ACTIVE CARIES	ARRESTED CARIES
Texture	Soft / leathery	Hard
Surface	Matte, rough	Shiny, smooth
Color	White, yellow, light or dark warm brown	Cool dark brown or black
Plaque	Present	Minimal/absent
Location	Protected (plaque stagnation)	Cleansable
Progression	Ongoing	Stopped
Risk Modifiers	Good saliva, hygiene, fluoride exposure, motivated towards behavior change	Xerostomia, anti-fluoride sentiment, recent history of carious treatment, high frequency cariogenic diet

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Why are arrested caries black?

- During the active phase, carious lesions are porous and pick up pigments (coffee, tea, wine, tobacco, iron) that become trapped after the surface hardens from remineralization
- Remineralization makes the surface harder, denser, and shinier. This alters light refraction, masking layers of underlying colors and contributing to darker, cooler appearance
- Breakdown products of collagen and protein can react with sugars forming pigmented complexes
 - o In food science, this is known as the Maillard reaction, which forms the sear on a steak or the crust on toast
- Reduced bacterial activity
 - o In an active lesion, the bacterial activity and acid production would dilute and displace the accumulation of pigment. Without that activity, the pigment persists.
- Silver Diamine Fluoride
 - o Covered in fluoride lecture.
 - o A form of fluoride often used to arrest or slow the growth of caries. Contains silver, which turns black on oxidization

ICCMS™ Code	Characteristics of Lesion	
	Signs of Active Lesions	Signs of Inactive Lesions
ICCMS™ Initial and Moderate Caries Stage Enamel	Surface of enamel is whitish/yellowish; opaque with loss of luster, feels rough when the tip of the ball-ended probe is moved gently across the surface. Lesion is in a plaque stagnation area, i.e. in the entrance of pits and fissures, near the gingival margin or, for proximal surfaces, below or above the contact point. The lesion may be covered by thick plaque prior to cleaning.	Surface of enamel is whitish, brownish or black. Enamel may be shiny and feels hard and smooth when the tip of the ball-ended probe is moved gently across the surface. For smooth surfaces, the caries lesion is typically located at some distance from the gingival margin. Lesion may not be covered by thick plaque prior to cleaning.
ICCMS™ Extensive Caries Stage Dentin	Dentin feels soft or leathery on gentle probing.	Dentin is shiny and hard on gentle probing.

Table 5. Characteristics of lesion activity across the ICCMS™ coronal caries stages.



Previously, caries was treated as as binary.
Cavity vs. No cavity.

Now we think of caries as a spectrum. The ICDAS rating system is an attempt to answer the question "where is this lesion on the disease spectrum?"

Once a scale is established, it becomes easier to have discussions and standardize the way we treat different lesion on the spectrum.

2002



2012





- A fundamental guidance statement relating to treatment decisions around operative intervention was agreed by all participants early in the development process and remains central to ICCMS™- this is to: **Preserve tooth structure and restore only when indicated.**
- Preservation of tooth structure in its widest sense drives all decisions in the ICCMS™, as a patient-centered and biologically compatible system which is evidence-based (within the limitations of current knowledge), preventively oriented and safe for tooth structure.

The ICDAS Foundation vision is to improve human health worldwide by the prevention and control of dental caries throughout life.

The goals of ICCMS™ are:

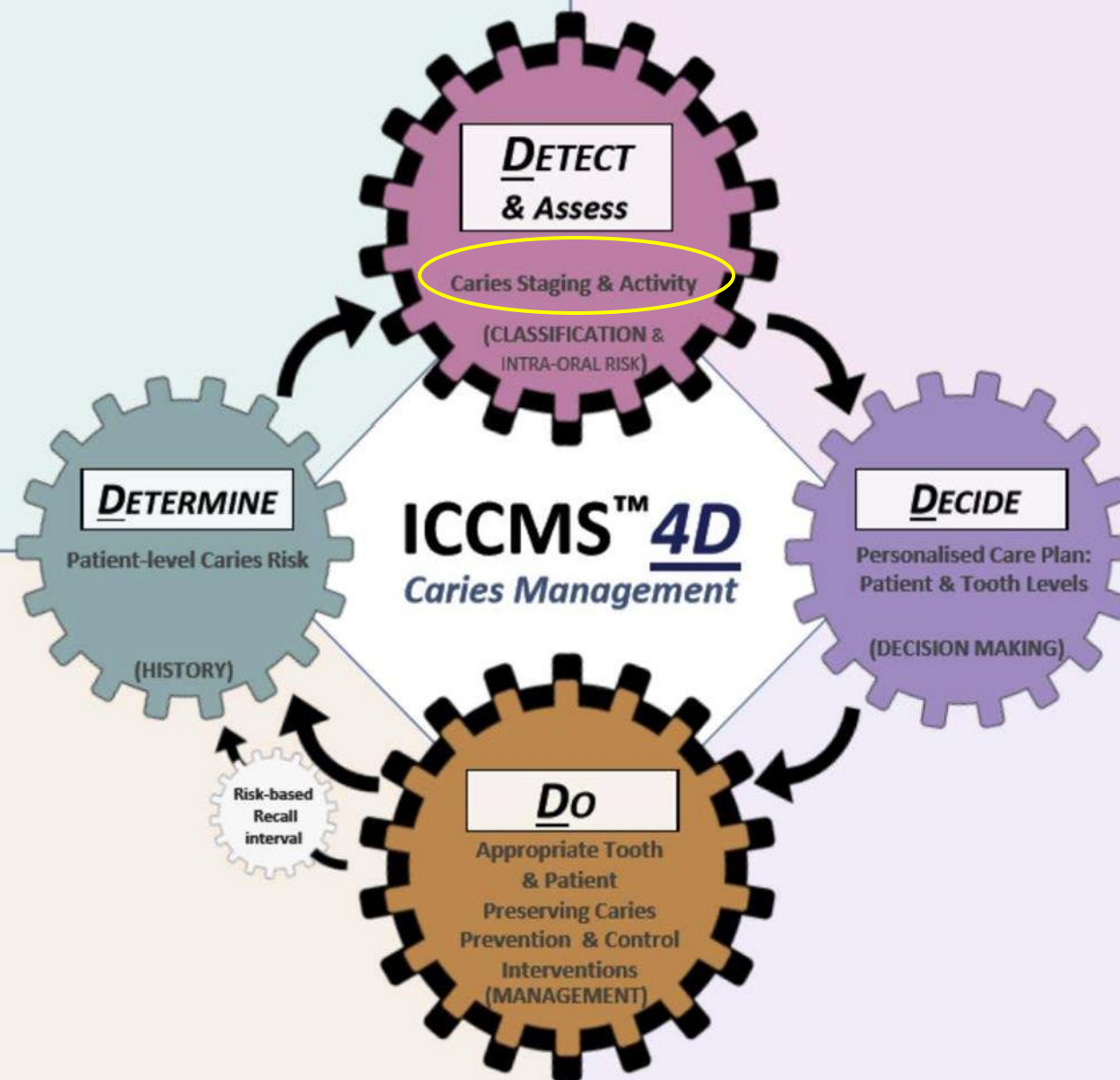
- to prevent new lesions from appearing
 - to prevent existing lesions from advancing further
 - to preserve tooth structure with:
 - * Non-operative care at more initial stages, and
 - * Conservative operative care at more extensive caries stages
- ...while managing risk factors and recalling patients at appropriate intervals, with periodic monitoring and reviewing.

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ICCMS™
CARIES MANAGEMENT
International Caries Classification and Management System

The International Caries Classification and Management System is a health outcomes focused system that aims to maintain health and preserve tooth structure. It uses a simple form of the ICDAS Caries Classification model to stage caries severity and assess lesion activity in order to derive an appropriate, personalised, preventively based, risk-adjusted, tooth preserving Management Plan.



Visual Exam

- **DRY and CLEAN teeth first**
 - **When you need to perform an exam prior to the completion of hygiene, leave a window open for re-evaluation after hygiene.**
 - **Diagnoses can change with experience, time, or factors outside the dentist's control. If you prioritize the doctor-patient relationship, your patient will understand.**
- Look everywhere systematically
 - Establish a pattern and stick to it. Don't get distracted by interesting things or the patient's CC.
 - Be suspicious of protected sites.
 - For EVERY tooth
 - If missing, say missing. Don't skip.
 - Existing (previous treatments)
 - Describe clinical findings (caries, non carious damage, abnormalities)
 - Correlate to radiographic findings
 - Consider other dental factors (perio, endo, occlusion, cosmetics, etc).
 - Make initial treatment recommendation

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If the teeth are heavily covered by plaque, it is necessary to clean the dentition before a proper caries diagnosis can be performed

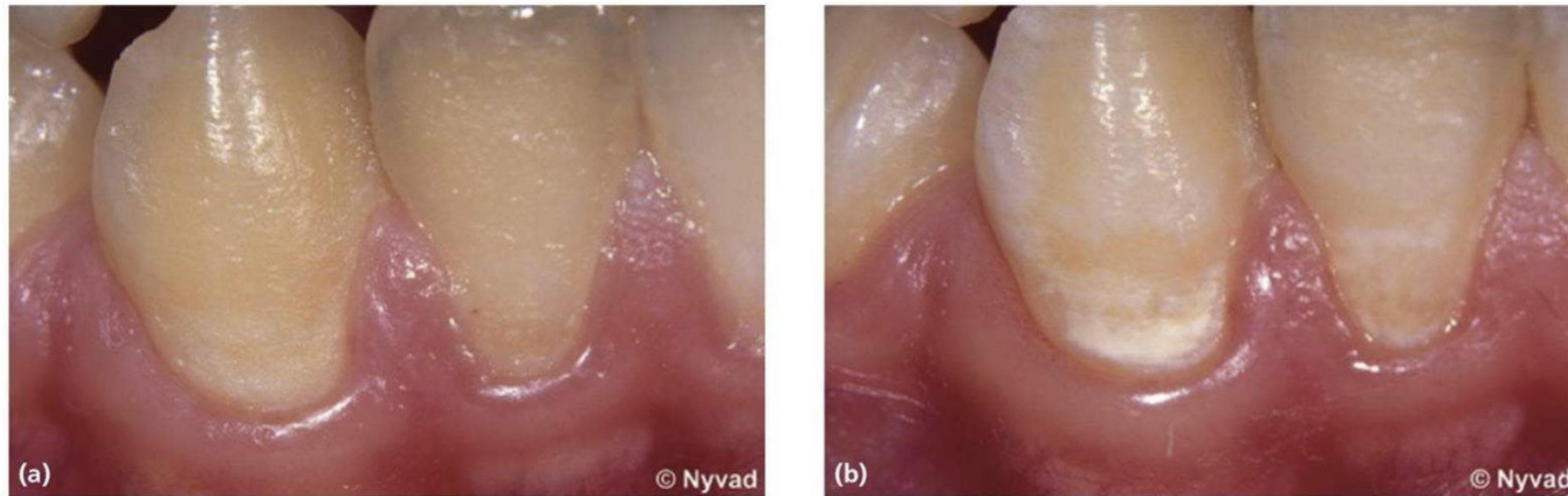


Figure 11.25 Lower canine and incisor (a) before and (b) after plaque removal. Note the presence of typical **active** noncavitated lesions after plaque has been removed with the side of a probe.

**How do we
know this
lesion was
active?**

Explorer Use?

- Some researchers are concerned that probing of suspected carious lesions may spread infective plaque (*S. Mutans*) to other teeth in the same mouth
 - o However, this concern has not been confirmed by longitudinal studies
 - o Furthermore, this hypothesis is incompatible with the ecological concept of caries. Transferred microorganisms would not survive or progress to caries unless their new eco niches favored it.
- What is happening when the tip of your explorer feels a "stick"?
 - o You're compressing and deforming demineralized enamel/dentin
 - o If 30-40% demineralization has occurred, you may cause cavitation
- Explore gently when necessary
 - o Carely drying and illumination often negate the need to explore
 - o Use other parts of the explorer to feel for texture



Figures 11.26–11.27 Figure 11.26: Examination of noncavitated caries lesion using the tip of a sharp probe that is moved gently across the surface of the lesion at an angle of 20–40° to assess lesion texture. Figure 11.27: Forceful poking with the probe perpendicular to the lesion should be avoided in order not to cause irreversible damage to the surface of the lesion!

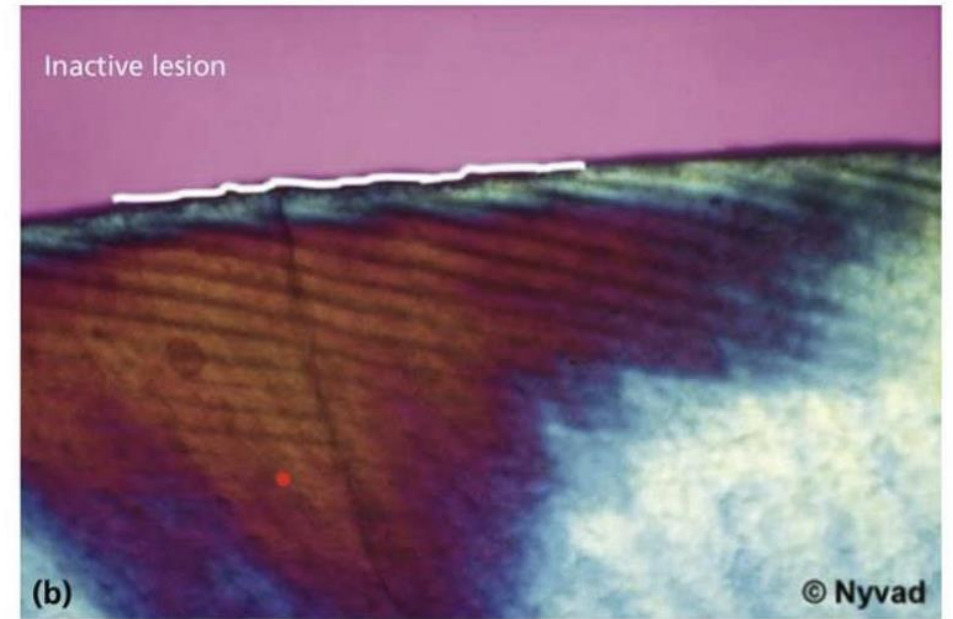
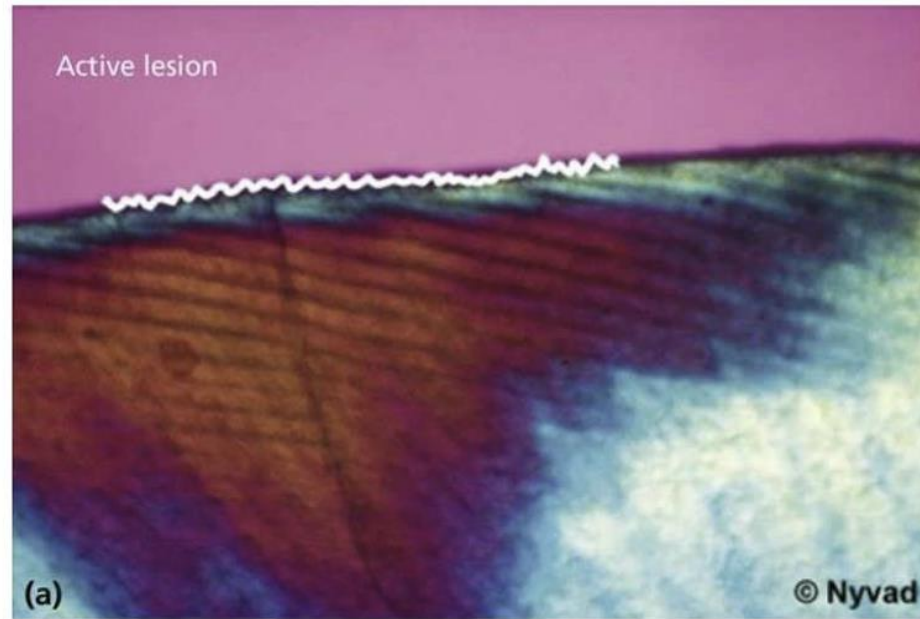
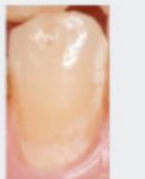
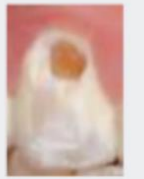
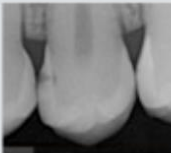
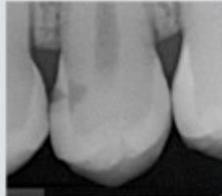


Figure 11.5 Polarized light microscopic images of noncavitated enamel caries lesion showing subsurface loss of mineral responsible for the opaque appearance of the lesion in the clinic. The white contour at the surface layer of the lesion indicates the principal difference between a rough/dull active lesion (a) and a smooth/shiny inactive lesion (b).

TABLE 2

American Dental Association Caries Classification System.

	AMERICAN DENTAL ASSOCIATION CARIES CLASSIFICATION SYSTEM						
	Sound	Initial		Moderate	Advanced		
Clinical Presentation	No clinically detectable lesion. Dental hard tissue appears normal in color, translucency, and gloss.	Earliest clinically detectable lesion compatible with mild demineralization. Lesion limited to enamel or to shallow demineralization of cementum/dentin. Mildest forms are detectable only after drying. When established and active, lesions may be white or brown and enamel has lost its normal gloss.		Visible signs of enamel breakdown or signs the dentin is moderately demineralized.	Enamel is fully cavitated and dentin is exposed. Dentin lesion is deeply/severely demineralized.		
Other Labels	No surface change or adequately restored	Visually noncavitated		Established, early cavitated, shallow cavitation, microcavitation	Spread/disseminated, late cavitated, deep cavitation		
Infected Dentin	None	Unlikely		Possible	Present		
Appearance of Occlusal Surfaces (Pit and Fissure)*,†	ICDAS 0 	ICDAS 1 	ICDAS 2 	ICDAS 3 	ICDAS 4 	ICDAS 5 	ICDAS 6 
Accessible Smooth Surfaces, Including Cervical and Root‡		 		 	 		
Radiographic Presentation of the Approximal Surface§	 E0¶ or RC0# No radiolucency	 E1¶ or RC1#  E2¶ or RC2#  D1¶ or RC3# Radiolucency may extend to the dentinoenamel junction or outer one-third of the dentin. Note: radiographs are not reliable for mild occlusal lesions.		 D2¶ or RC4# Radiolucency extends into the middle one-third of the dentin		 D3¶ or RC5# Radiolucency extends into the inner one-third of the dentin	

* Photographs of extracted teeth illustrate examples of pit-and-fissure caries.

† The ICDAS notation system links the clinical visual appearance of occlusal caries lesions with the histologically determined degree of dentinal penetration using the evidence collated and published by the ICDAS Foundation over the last decade; ICDAS also has a menu of options, including 3 levels of caries lesion classification, radiographic scoring and an integrated, risk-based caries management system ICCMS. (Pitts NB, Ekstrand KR. International Caries Detection and Assessment System [ICDAS] and its International Caries Classification and Management System [ICCMS]: Methods for staging of the caries process and enabling dentists to manage caries. *Community Dent Oral Epidemiol* 2013;41[1]:e41-e52. Pitts NB, Ismail AI, Martignon S, Ekstrand K, Douglas GAV, Longbottom C. ICCMS Guide for Practitioners and Educators. Available at: https://www.icdas.org/uploads/ICCMS-Guide_Full_Guide_US.pdf. Accessed April 13, 2015.)

‡ "Cervical and root" includes any smooth surface lesion above or below the anatomical crown that is accessible through direct visual/tactile examination.

§ Simulated radiographic images.

¶ E0-E2, D1-D3 notation system.³³

R0, RA1-RA3, RB4, and RC5-RC6 ICCMS radiographic scoring system (RC6 = into pulp). (Pitts NB, Ismail AI, Martignon S, Ekstrand K, Douglas GAV, Longbottom C. ICCMS Guide for Practitioners and Educators. Available at: https://www.icdas.org/uploads/ICCMS-Guide_Full_Guide_US.pdf. Accessed April 13, 2015.)

Caries Overview 2021: Epidemiology, Etiology, and Evidence-Into-Action Considerations

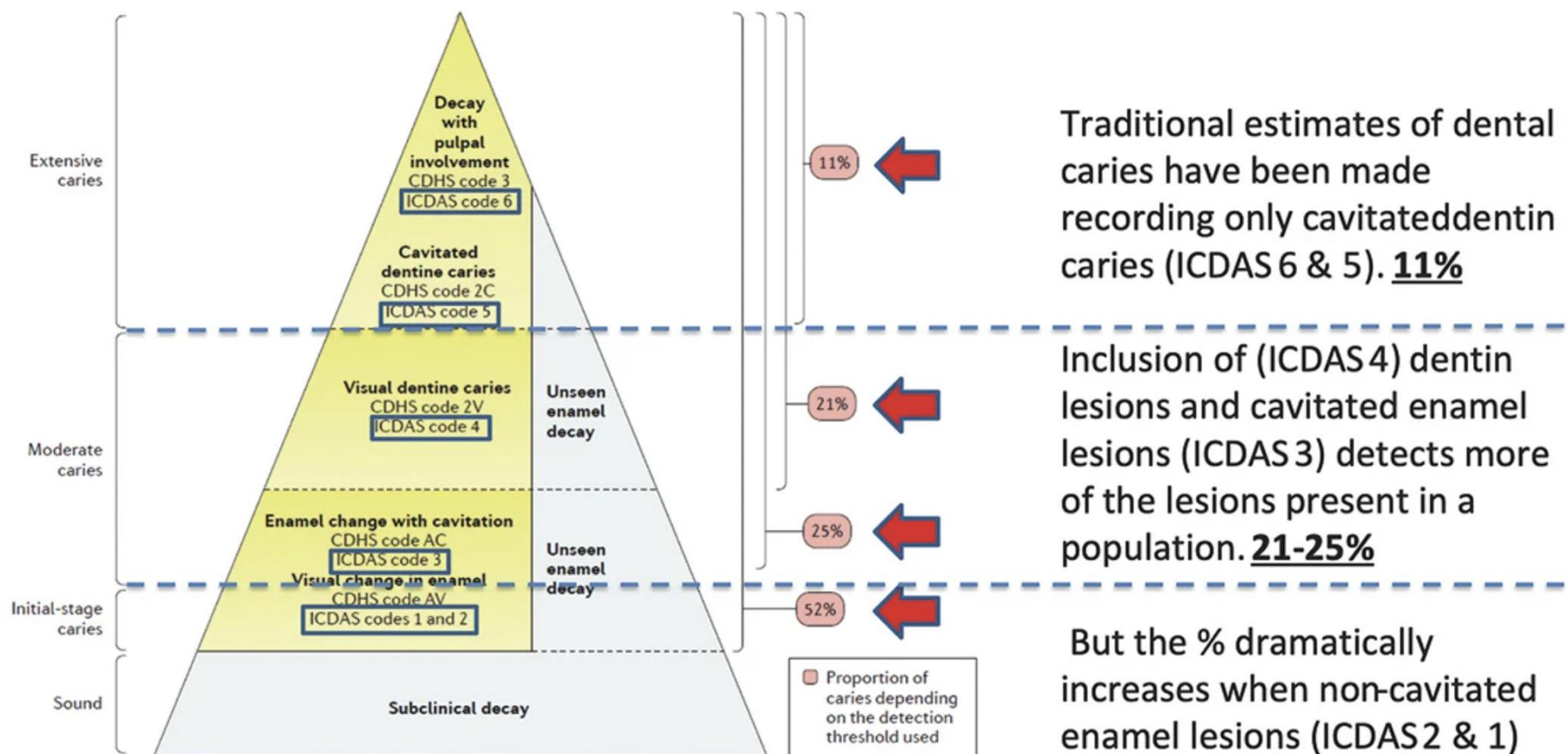
NIGEL PITTS, BDS, PHD, FRSE

Figure 1

Dental Caries Epidemiology: Impact of detection thresholds, 15 yr olds

The prevalence of initial stage-carries across populations is high and often not quantified

(Data on 15 year old children from 2013 National Child Dental Health Survey (CDHS) of England, Wales & NI)



Definition of ICCMS™ Caries Merged categories			
categories	Sound surfaces (ICDAS™ code 0)	 	Sound tooth surfaces show no evidence of visible caries (no or questionable change in enamel translucency) when viewed clean and after prolonged air-drying (5 seconds). 8-9 <i>(Surfaces with developmental defects such as enamel hypomineralization (including fluorosis), tooth wear (attrition, abrasion and erosion), and extrinsic or intrinsic stains will be recorded as sound).</i>
	Initial stage caries (ICDAS™ codes 1 and 2)	 	First or distinct visual changes in enamel seen as a carious opacity or visible discoloration (white spot lesion and/or brown carious discoloration) not consistent with clinical appearance of sound enamel (ICDAS™ code 1 or 2) and which show no evidence of surface breakdown or underlying dentin shadowing.

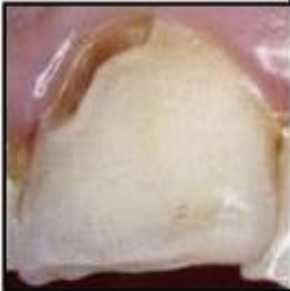
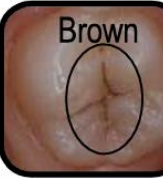
Caries	Moderate stage caries (ICDAS™ codes 3 and 4)	  A white or brown spot lesion with Localized enamel breakdown, without visible dentin exposure (ICDAS™ code 3), or an Underlying dentin shadow (ICDAS™ code 4), which obviously originated on the surface being evaluated. <i>(To confirm enamel breakdown, a WHO/CPI/PSR ball-end probe can be used gently across the tooth area - a limited discontinuity is detected if the ball drops into the enamel micro-cavity/discontinuity).</i>
	Extensive stage caries (ICDAS™ codes 5 and 6)	  A distinct cavity in opaque or discolored enamel with visible dentin (ICDAS™ code 5 or 6). <i>(A WHO/CPI/PSR probe can confirm the cavity extends into dentin).</i>

Table 2. Definition of ICCMS™ Caries categories (merged codes).

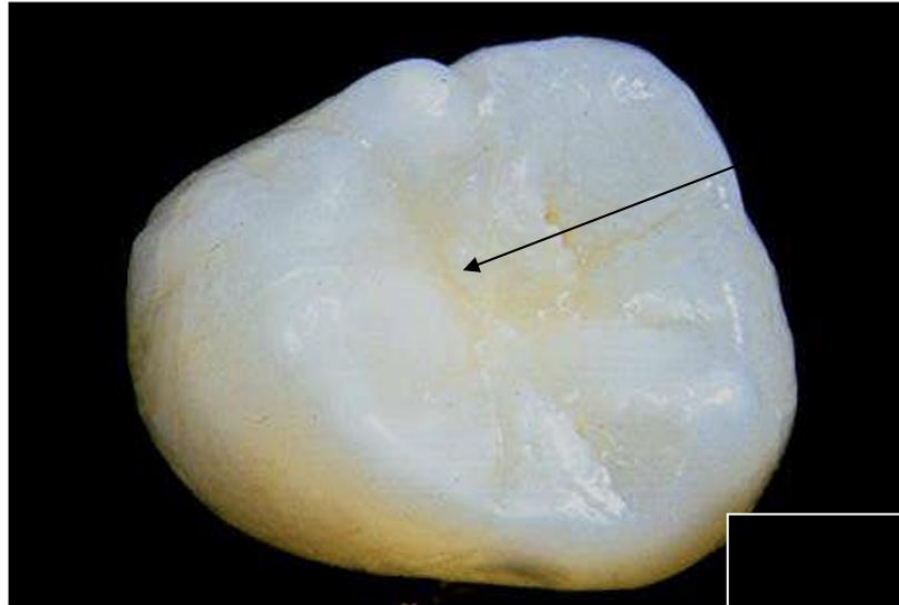
ICDAS scores	Smooth surfaces	Occlusal surfaces
0 Sound		
1 First Visual Change in Enamel <i>when dry</i>		 
2 Distinct Visual Change in Enamel <i>when wet</i>		 
3 Localized Enamel Breakdown		
4 Underlying Dark Shadow from Dentine		
5 Distinct Cavity with Visible Dentine		
6 Extensive Distinct Cavity with Visible Dentine		

Considering
crown/RCT

1, service, scholarship and social

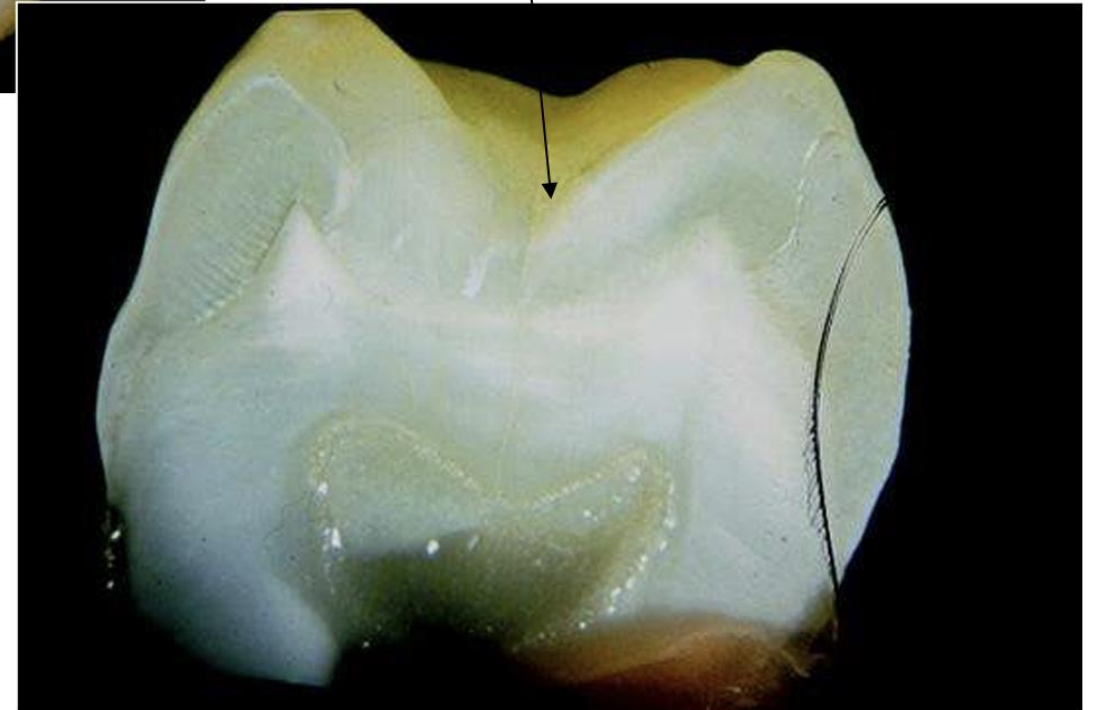
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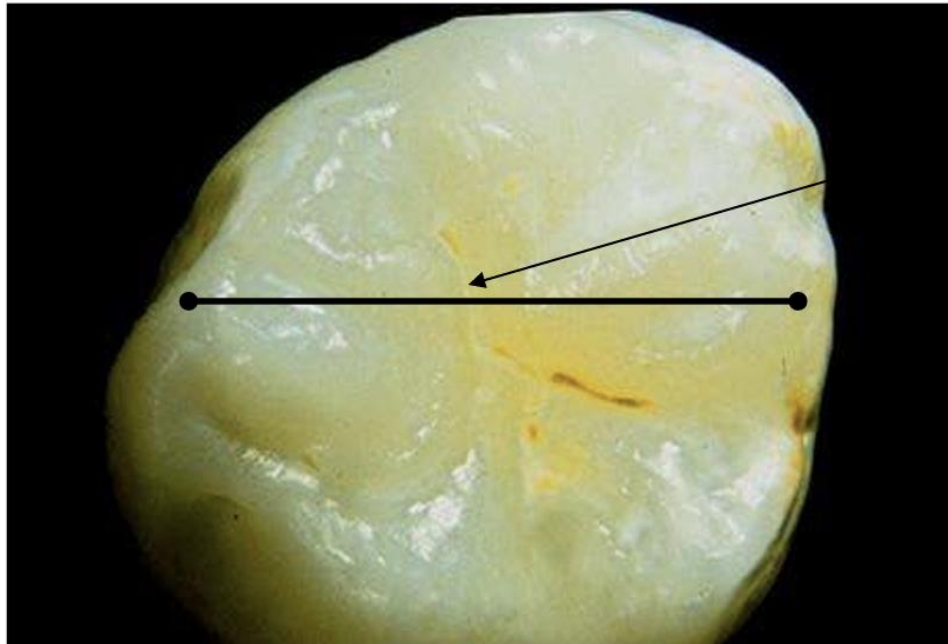
After 5 seconds of
drying, enamel
appears sound.



ICDAS score 0

Histological score 0



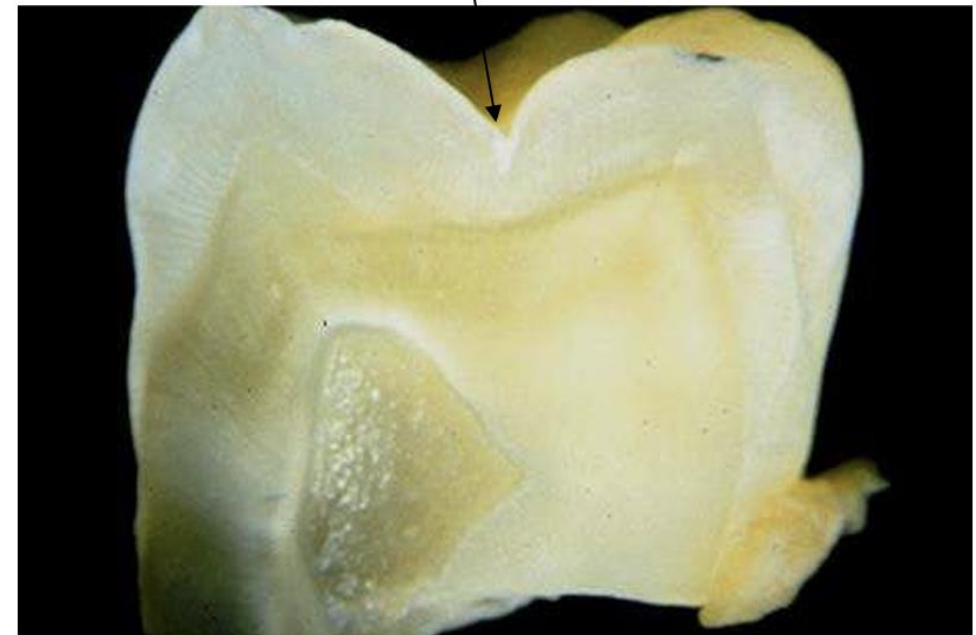


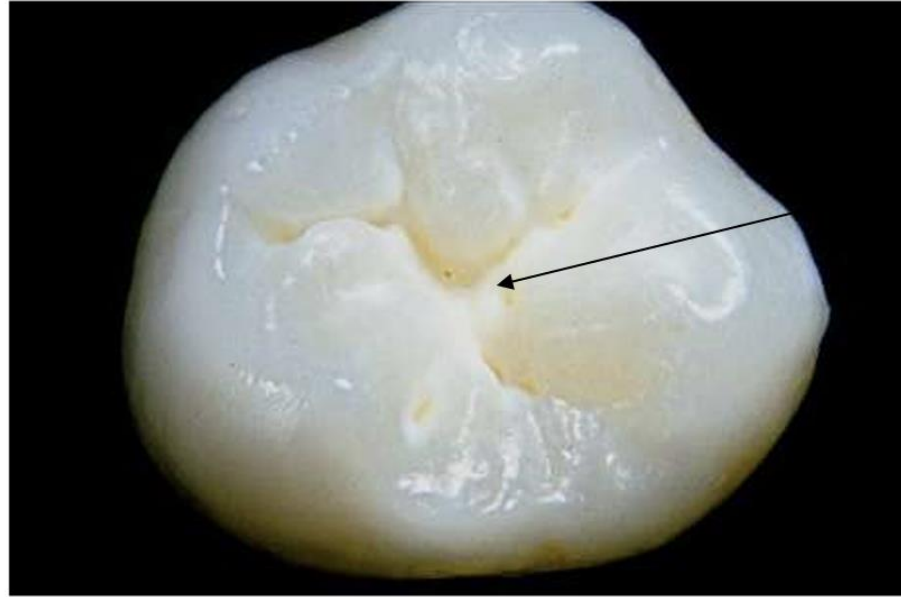
ICDAS score 1

Histological score 1

After 5 seconds of drying, there are distinct visual changes in the enamel (WSL or brown discoloration)

Histologically, there is E1 demineralization.





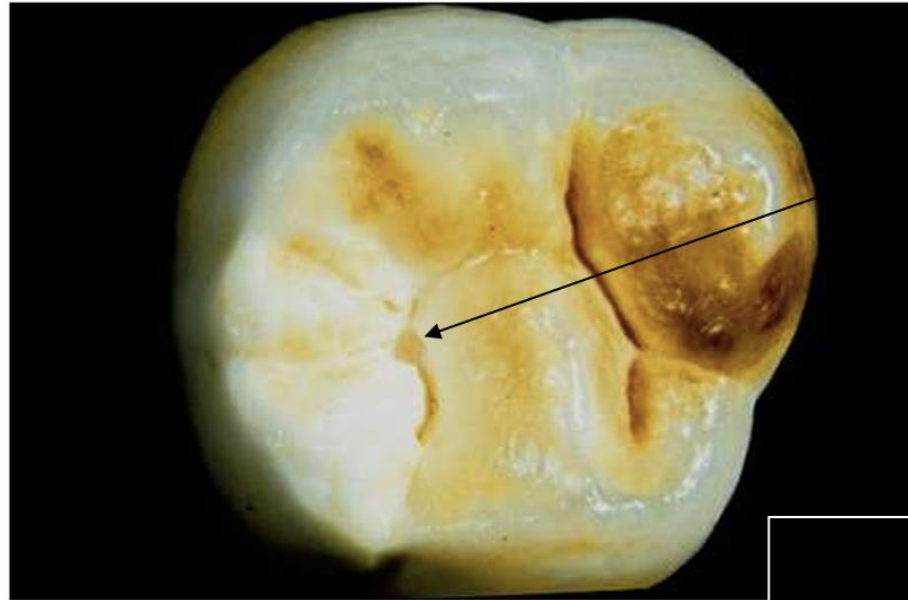
ICDAS score 2

Histological score 2



EVEN WHEN WET, there are distinct visual changes in the enamel (WSL or brown discoloration).
Lesion may be wider than the natural fissure or contact area.

Histologically, there is E2-D1 demineralization.



ICDAS score 3

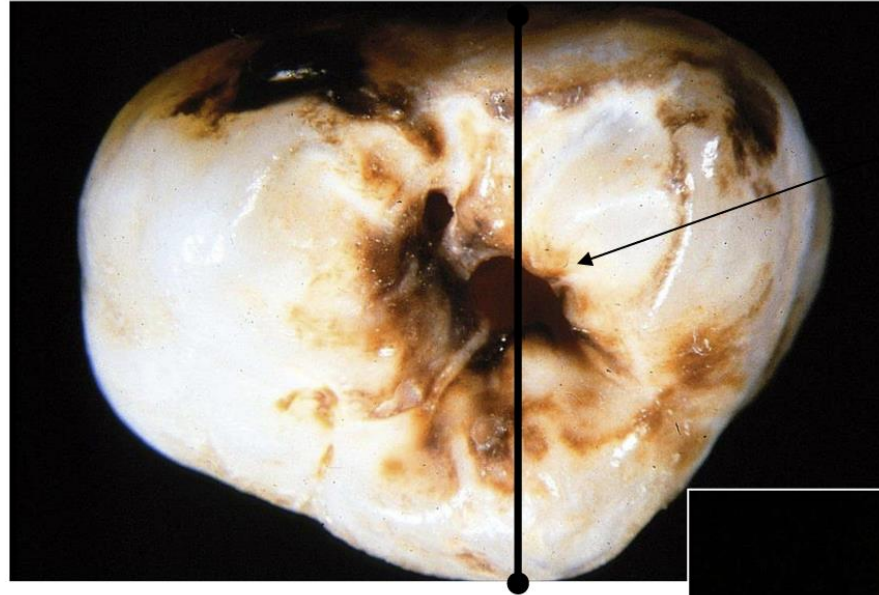
Histological score 3

3: Localized enamel breakdown is visible (microcavitation). But there is no **visible** dentin involvement yet.

4: Underlying **dentinal shadowing** is visible.

Histologically, there is D2 demineralization in both a ICDAS 3 and 4.





ICDAS score 5

Histological score 4

5: Clear enamel cavity with dentin exposed and visible. Lesion is localized to one area.

6: Larger and deeper. More than half of the surface destroyed or cusp undermined. Structural integrity is compromised.

Histologically, D3 demineralization.



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Root Caries ICDAS Categorization and description

Code 0 (Sound)	The root surface does not exhibit any unusual discoloration that distinguishes it from the surrounding or adjacent root areas nor does it exhibit a surface defect either at the cement-enamel junction or wholly on the root surface. The root surface has a natural anatomical contour, <u>or</u> may exhibit a definite loss of surface continuity or anatomical contour that is <u>not</u> consistent with the dental caries process. This loss of surface integrity usually is associated with dietary influences or non-carious lesions such as abrasion or erosion. These conditions usually occur on the facial (labial) surface. These areas are typically smooth, shiny and hard. Abrasion is characterized by a clearly defined outline with a sharp border, whereas erosion has a more diffuse border. Neither condition shows discoloration.
Code 1 (Initial lesion)	There is a clearly demarcated area on the root surface or at the cement-enamel junction (CEJ) that is discolored (light/dark brown, black) but there is no cavitation present (loss of anatomical contour < 0.5 mm).
Code 2 (Moderate/ Extensive lesion)	There is a clearly demarcated area on the root surface or at the CEJ that is discolored (light/dark brown, black) and there is cavitation (loss of anatomical contour ≥ 0.5 mm $\leq$ -2 mm (Moderate), > 2mm (Extensive)) present.

The characteristics of the base of the discolored area on the root surface can be used to determine whether the root caries lesion is active or not. These characteristics include appearance (shiny/glossy or matte/non-glossy), location in a plaque-stagnation area/not, and perception of texture on gentle probing (soft / leathery, or hard / rough / smooth). Active root caries lesions usually located within 1 mm of the crest of the gingival margin [Ekstrand et al, 2008, 2013].

The following diagram of characteristics of lesion activity across the ICDAS root caries stages may be useful for reference when examining for root caries:

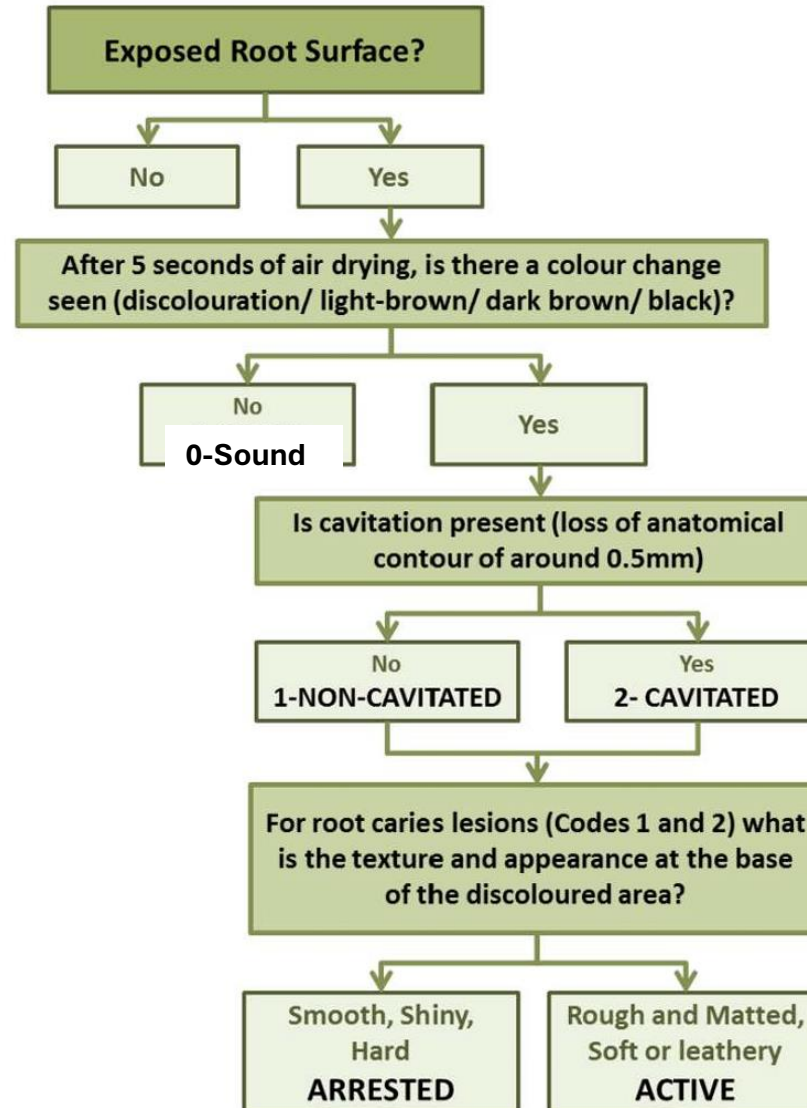




Figure 11.7 (a) Active root surface caries lesion in upper canine presenting a softened surface. (b) Inactive root surface caries lesion in upper incisor showing a smooth discolored surface.

The management of root caries is most based on expert opinions. The evidence base for supporting any management strategy is either weak or does not exist. ICCMS™ recommends the following management strategies^{49,60,61,164-168}.

ICDAS Root Categories	Activity status	
	Active (leathery and/or soft)	Inactive (hard and shiny)
Root Sound	Risk-based Prevention (Refer to Previous Table)	
Root Initial (Non-cavitated)	Oral hygiene, fluoride varnish, saliva stimulation, high fluoride toothpaste*; chlorhexidine varnish	Refer to Primary Prevention Table and Emerging Evidence section
Root Moderate/Extensive (Cavitated)	TPOC (Glass ionomer cements, composite resin) Oral hygiene, fluoride varnish*, saliva stimulation, high fluoride toothpaste; chlorhexidine varnish	Refer to Primary Prevention Table and Emerging Evidence section TPOC if there is sensitivity or for esthetic reasons

* Best evidence is for the efficacy of 5% fluoride varnish rather than lower fluoride level varnishes.



Detect and Assess: Caries Staging and Activity

- Caries vs Not Caries?
- Active vs. Inactive?
- Cavitated vs Not Cavitated?
- ICDAS rating?

- Other types of classification?

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